

Lab 3: Architectural Style Selection and Justification

Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Architectural Choice and Reasons

The system is divided into separate layers/components for user interaction, business processing, and data access. The **Kiosk UI** handles touchscreen interactions, while the **Order Manager** handles order processing and coordinates payment and receipt generation. The **Menu & Pricing Database** stores coffee and pricing information.

Reason 1 – Clear separation of concerns: Layered Architecture separates presentation, business logic, and data-related responsibilities. This makes the kiosk easier to understand, maintain, test, and modify. For example, changes to coffee prices can be handled in the menu/data component without changing the order-processing logic.

Reason 2 – Suitable for a relatively simple kiosk system: The coffee kiosk has a straightforward workflow involving menu selection, order management, payment, and receipt printing. A layered structure provides the required organization without introducing the operational complexity of a distributed architecture.

Security Advantage

The architecture separates the user interface from payment processing and data storage. Payment-related operations can therefore be handled by the dedicated **Payment Service**, reducing direct exposure of payment processing logic to the touchscreen interface and allowing access between components to be controlled through defined interfaces.

Performance Benefit

Separating responsibilities allows frequently accessed menu and pricing information to be handled independently from payment and receipt operations. This can reduce unnecessary processing in the user-interface layer and keep the ordering interaction responsive.